Thermal oxidation

Part A. Theoretical principles of ellipsometry (4.5 points)

A.1 (0.2pt)

A.2 (0.4pt)

$k_{1x} =$

$k_{2x} =$

$k_{3x} =$

A.3 (0.6pt)

Equation:

A.4 (0.6pt)

$r_s =$

$A =$ $B =$

A.5 (0.6pt)

Equation:

A.6 (0.6pt)

$r_p =$

$A =$ $B =$

A.7 (1.5pt)

$\theta_{\text{opt}} =$

Part B. Kinetics of oxidation (5.5 points)

B.1 (0.7pt)

$\Phi =$ $d =$

B.2 (0.7pt)

$\Phi =$ $d =$

B.3 (0.3pt)

t, min	d, nm

B.4 (0.3pt)

$v =$

B.5 (0.5pt)

$C_0 =$

$C_i =$

B.6 (0.5pt)

Equation:

$A =$ $B =$

$\tau =$

B.7 (1.0pt)

T	B	B/A

B.8 (1.5pt)

$\Delta E_b =$ $\Delta E_a =$

Plot:

B

B/A